Lesson 1 · Passphrases you can remember · for your pupils

Passphrase builders

40 minutes | Ages 10–16 · adapt for other ages below | No electricity needed | No devices needed | Photocopy-friendly

Big question

What makes a password hard to guess but easy to remember?

Pupils will be able to…

- Explain why three random words are harder to guess than a name and a birthday.
- Build a practice passphrase from three random word cards and remember it.
- Name three things they never share with friends or callers: a password, a PIN and a one-time code.

You need

- Board and chalk (or paper on the wall)
- About 40 word cards on scrap paper, in a bag or tin
- A real padlock, or a drawing of one with number dials
- Worksheet: Password power (one per pair, optional)

Before class

Write about 40 everyday nouns on small cards — mango, bucket, rocket, drum, sandal, cloud — and mix them in a bag. On the board, write “Musa2012” and draw a padlock with three number dials.

Time	What you do	What you can say	Pupils
Warm-up 0–5 min	Show the padlock with three number dials. Ask how many combinations a thief would need to try, then how many with six dials. Write 1,000 and 1,000,000 on the board.	*“Each extra dial makes the thief’s job much longer. Passwords work the same way — every extra letter adds work.”*	Guess the number of tries; notice that longer means many more.
Teach 5–13 min	Point to “Musa2012”. Explain that guessing programs try names, birthdays and common tricks first. Pull three cards from the bag, write the passphrase, and draw one funny picture that links the words.	*“A name and a year are easy to guess. Three random words are long and surprising — and a funny picture helps you remember.”*	Say why “Musa2012” is weak; repeat the three-word passphrase.
Word cards 13–25 min	Pairs pull three cards, write a practice passphrase and draw one picture linking the words. Cards go back in the bag. At the end of the lesson, pairs try to recall their words from the picture alone.	*“Make it random — no names, birthdays or football teams. This is practice: real passphrases stay secret.”*	Build a practice passphrase and draw a memory picture.
Pair talk 25–33 min	Read out four situations: a friend asks for your password; a caller asks for a code sent to a parent’s phone; one password for everything; a list kept safe at home. Pairs decide “safe” or “not safe” and say why.	*“Never give a password, PIN or code to a friend, a caller or a message. If you’re unsure, ask a trusted adult.”*	Decide safe or not safe; give a reason.
Wrap-up 33–40 min	Pairs look at their picture and try to recall their three words. Then exit ticket: one rule for a strong passphrase and one thing they never share.	*“Tonight, explain the three-random-words idea to someone at home — but keep any real passphrase secret.”*	Recall their words; complete the exit ticket.

Check understanding
Listen for “long” and “random”, not “add more symbols”. If a pupil picks a name or a birthday, ask: “Could someone who knows you guess this?”

Exit ticket
One rule for a strong passphrase + one thing I never share.

Common mix-ups

Pupils may think…	Help them see…
More symbols always make a password strong.	Show “P@ssw0rd” — guessing programs know these swaps. Length and randomness matter more.
My best friend can know my password.	Friendships change and messages get forwarded. Keep passwords, PINs and codes to yourself; ask a trusted adult if unsure.
A famous saying makes a good passphrase.	Famous sayings and song lines are easy to guess. Pick words at random, like the cards in the bag.

Adapt it

Younger pupils
Ages 6–9: use picture cards and make a silly three-picture story. Focus on one rule: passwords are secrets you share only with a parent or carer.

Older pupils and university
Work out the combinations for a 4-digit PIN (10,000) and a 6-digit PIN (1,000,000). Discuss why sites should block passwords that have already leaked.

Large classes
Give each row a bag of cards. Row leaders read one practice passphrase aloud; the class shows thumbs up for random, thumbs down for guessable.

No board or no space
Write the word cards on leaves, stones or scraps of paper and lay them in a circle outside. Pupils walk round and pick three.

Short on time? Do the padlock warm-up, the teach step and one round of word cards (20 minutes).

What you need to know

A passphrase of three random words is easier to remember and harder to guess than a name, a date and a symbol. Keep email and mobile money passwords different; never share them.

Passphrase
A password made of several words, such as three random words joined together. It is long and hard to guess, but easy to picture and remember.

PIN
Personal identification number: a short number code that protects a phone, a bank card or a mobile money account. Only you should know it.

One-time code
A short code sent to your phone to prove it is really you. It works once, for a short time. Never give it to anyone.

Reusing a password
Using the same password for more than one account. If one website is hacked, criminals try that password on your other accounts too.

My notes — how did it go?